

# Aliah University

End-Semester Examination (Spring Semester) – 2024-25

(1<sup>st</sup> Year 2<sup>nd</sup> Semester 2-yrs MCA)

Subject Name: Data structures and Algorithms  
Subject Code: MCAPGPC08

Full Marks: 80  
Time: 3hrs

## Group-A (Answer all questions)

5 × 2 = 10

1. What is an Abstract Data type (ADT)? Explain it with example. [CO1, Remembering]
2. For the given infix notation  $M+H-A*B-H/K$ , find the prefix and post fix notation. [CO3, Understanding]
3. Define complete binary tree with an example? [CO1, Remembering]
4. Define dequeuer with an example. [CO1, Remembering]
5. Discuss the time complexities of Bubble sort. [CO1, Remembering]

## Group-B (Answer any six questions)

6 × 5 = 30

1. What is data structure? Why do we need data structure? Differentiate between linear and non-linear data structure? [CO1, Remembering] 1 + 2 + 2
2. Write a C program to show the parameters passing by reference in a C function. [CO3, Understanding] 5
3. Represent the following sparse matrix using link list. [CO3, Understanding] 5

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 0 | 8 | 0 | 0 | 0 |
| 0 | 7 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 5 |
| 0 | 3 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 |

4. What is recursion? Write a recursion algorithm for finding factorial of a number. [CO3, Understanding] 1 + 4
5. Sort the following elements [6, 3, 5, 8, 1, 4, 2, 7] using quick sort. [CO3, Understanding] 5
6. Evaluate the following prefix expression  $[-++134/427]$  using stack. [CO4, Applying] 5
7. The pre-order and in-order traversals of a tree yield the following sequence of nodes: [CO3, Understanding] 5
  - Pre-order: A B D E G H C F
  - In-order: D B G E H A C F

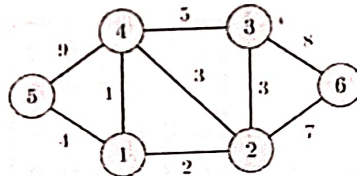
Draw the binary tree  $T$  and find out its Post-order sequence.

8. What are the limitations of queue? Explain with suitable example the insertion operation in a circular queue.  
[CO3, Understanding] 1 + 4

**Group-C**  
(Answer any four questions)

$4 \times 10 = 40$

1. What is linked list? Write and explain the algorithm for creation of a node and insertion of a node at the beginning of the singly list.  
[CO3, Understanding] 1 + 4 + 5
2. Consider the following array of integer values [23, 12, 35, -3, 4, 41, 10, 17, 5]  
[CO4, Applying] 5 + 5
- a. Write the contents of the array after 5 passes of the outermost loop of insertion sort.
- b. Write the contents of the array after 5 passes of the outermost loop of selection sort.
3. What is queue? What are the operations of queue? Write an algorithm to insert an element into a queue.  
[CO3, Understanding] 1 + 2 + 7
4. Construct a binary search tree using the following elements [11, 2, 14, 5, 1, 7, 4, 15] and then delete the following elements one by one [11, 2, 14].  
[CO4, Applying] 5 + 5
5. What is stack? Write algorithm for operations of stack with examples.  
[CO3, Understanding] 1 + 9
6. Find out the minimum spanning tree of the following graph step by step using Prim's and kruskal's algorithms.  
[CO4, Applying] 5 + 5



## End-Semester Examination (Spring Semester) – 2024-25

[MCA – 1<sup>st</sup> Year, 2<sup>nd</sup> Semester]

Paper Name: Object Oriented Programming with Java  
Paper Code: MCAPGPC11

Full Marks: 80  
Time: 3 Hrs

## GROUP A

[Answer any five questions]

[5 x 2 = 10]

1. i) Mention any two benefits of exception handling.  
ii) What do you mean by method signature?  
iii) Define the term “compile-time error” with an example.  
iv) Explain the purpose of ‘throws’ keyword.  
v) How does Java achieve platform independence?  
vi) Explain the use of the join () method in the context of multithreading in Java.

## GROUP – B

[Answer any four questions]

[4 x 5 = 20]

2. i) What are the conventions for naming classes in Java?  
ii) Differentiate compile-time polymorphism from dynamic polymorphism. [2+3]  
iii) Explain the usage differences among instance variables, local variables, and static variables with examples.  
iv) Briefly state the fundamental properties of object-oriented programming languages.  
v) Create an interface Area with the following details:  
Data Member: a) float PI = 3.14159F  
Methods: a) float computeArea (float a, float b)  
Implement the above interface in two different classes, Cylinder and Rhombus.  
vi) Write a Java program which can simultaneously run two threads with appropriate priorities after creating the threads by extending the Thread class.

## GROUP – C

[Answer any five questions]

[5 x 10 = 50]

7. a) What is a constructor and what are the advantages of using it?  
b) What are the purposes of ‘this’ keyword in Java?  
c) Write a program to find the Area and Volume of a Cone by extending a Shape Class which contain an abstract method area () [Note: the volume () method need not to be defined in the Shape class]. [3+2+5]  
8. a) What is the purpose of the finalize() method in Java?  
b) What are the different types of inheritances? Explain with examples.  
c) What do you mean by garbage collection?  
d) What is the base class of all types of Exception? [2+5+2+1]

9. a) What are the different access modifiers available in Java? State their access scope.  
b) Explain try-catch-finally block with a real-time example. [(1+4)+5]
10. a) Explain the significances of the keywords in the statement public static void main (String args[]).  
b) How an abstract class is different from an interface?  
c) With an example differentiate pre- and post- increment operators. [4+3+3]
11. a) Explain the different types of shift operators in Java with proper example. What is ternary operator?  
b) What are the advantages of organizing classes into packages?  
c) Draw and explain the state transition diagram of a Thread. [3+1+2+4]
12. Explain the 'final', 'static', and 'super' keywords with all their usages. Support explanation with a suitable Java program. [5+5]

# Aliah University

## Spring Semester Examination - 2025

### MCA (CSE) 1st year, 2nd Semester Examination

Paper Name: Artificial Intelligence

Paper Code: MCAPGPC12

Full Marks: 80

Time: 3 hr

| 1.                                       | Group-A (Answer all questions)                                                                                                                                                               | 2 x 5 = 10 |                 |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------|
| a.                                       | Differentiate between an agent and its environment in AI systems.                                                                                                                            | 2          | (CO1, Rem)      |
| b.                                       | Given raw input features [200, 500, 1000], scale them to the range [-1, 1] using min-max normalization.                                                                                      | 2          | (CO1, Eval)     |
| c.                                       | Consider the two crisp sets A and B as given below. $A = \{1, 4, 6, 8, 9\}$ $B = \{11, 12, 13, 14\}$ . Find the relation matrix R with $R = \{(a, b)   b = a + 2, (a, b) \in A \times B\}$ . | 2          | (CO2, App)      |
| d.                                       | What is the T-norm operator?                                                                                                                                                                 | 2          | (CO3, Rem)      |
| e.                                       | What are composite Fuzzy relationships?                                                                                                                                                      | 2          | (CO3, Rem)      |
| Group-B (Answer any Five (05) questions) |                                                                                                                                                                                              | 5 x 6 = 30 |                 |
| 2.                                       | Differentiate classification and clustering techniques and explain their usabilities? What is the learning rate?                                                                             | 2+2<br>+2  | (CO1, Und)      |
| 3.                                       | Critically evaluate the role of hyperparameter tuning in improving performance. Write short notes on Activation functions                                                                    | 2+4        | (CO2, Eval)     |
| 4.                                       | Explain the triangular, trapezoidal, and Gaussian fuzzy membership functions mathematically. What do you mean by fitness function in GA?                                                     | 5+1        | (CO3, Rem)      |
| 5.                                       | Discuss with proper examples for Fuzzy inference systems. Discuss the fuzzy c-means algorithm.                                                                                               | 2+4        | (CO3, Rem)      |
| 6.                                       | Compare Decision Trees and Random Forest algorithms with respect to accuracy and interpretability. What is the meaning of convergence in neural network training?                            | 4+2        | (CO3 Ana, Eval) |
| 7.                                       | Differentiate Mamdani and Larsen rules of Fuzzy relationships. Define a linguistic variable with an example.                                                                                 | 5+1        | (CO4 App, Ana)  |

| Group-C (Answer any Four (04) questions) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4 x 10 = 40 |                  |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------|
| 8.                                       | <p>Solve the classic XOR problem using a two-layer perceptron network. Show the network architecture and activation functions used. Draw and explain the structure of a typical intelligent agent. Given membership functions <math>\mu_A(x) = 0.6</math> and <math>\mu_B(x) = 0.8</math>, compute the fuzzy implication using Zadeh's implication rule: <math>\mu_{A \rightarrow B}(x) = \max(1 - \mu_A(x), \mu_B(x))</math>.</p>                                                                                                                                                                                                                    | 6+4         | (CO2, App & Ana) |
| 9.                                       | <p>K-Means Clustering Problem Given data points: (1, 2), (1, 4), (5, 8), (6, 8). Apply one iteration of K-Means clustering with <math>K = 2</math> and initial centroids (1, 2) and (5, 8). Discuss any three ML classifiers with suitable examples.</p>                                                                                                                                                                                                                                                                                                                                                                                              | 4+6         | (CO2, App & Ana) |
| 10.                                      | <p>If temperature is High then fan speed is Fast else fan speed is Slow. Represent this fuzzy rule using implication <math>A \rightarrow B</math>. Define suitable fuzzy sets for High temperature and Fast fan speed. Find R, where 'slow to high' fuzzy values will be from <math>[0,1]</math>. What are Zadeh's arithmetic and max-min rules in Fuzzy systems?</p>                                                                                                                                                                                                                                                                                 | 5+5         | (CO3, App & Ana) |
| 11.                                      | <p>If <math>X = (x_1, x_2, x_3)</math>; <math>Y = (y_1, y_2)</math>; <math>Z = (z_1, z_2, z_3)</math>; Then find <math>X \circ Z</math> relation matrix using</p> $R = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \end{matrix} \quad S = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.4 & 0.7 \\ 0.5 & 0.8 & 0.9 \end{bmatrix} \end{matrix}$ <p>Distinguish Fuzzy relations, Fuzzy implications and Fuzzy inference systems.</p> | 3+7         | (CO4, App & Ana) |
| 12.                                      | <p>In a simple 2-layer neural network, the output neuron receives input <math>z = 0.5</math>. The actual output is <math>y = 0.6</math> and the target is <math>t = 1</math>. Compute the error and gradient using mean squared error loss <math>E = 1/2 * (t - y)^2</math>. Given <math>X = \{a, b, c, d\}</math>, <math>Y = \{1, 2, 3, 4\}</math>, whereas <math>A = \{(a,0.0), (b,0.8), (c,0.6), (d,1.0)\}</math> &amp; <math>B = \{(1,0.2), (2,1.0), (3,0.8), (4,0.0)\}</math>. Apply Zadeh's min-max rule to determine the implication relation: If <math>x</math> is <math>A</math> then <math>y</math> is <math>B</math>.</p>                  | 4+6         | (CO3, App & Ana) |

**Aliah University**  
**End-Semester Examination (Spring Semester) - 2025**  
**(For 1<sup>st</sup> Year 2<sup>nd</sup> Semester MCA (CSE))**

**Paper Name: Computer Networks**  
**Paper Code: MCAPGPC10**

**Full Marks: 80**  
**Time: 3 hrs**

**Group-A**  
**(Answer *all* questions)**

**10 × 1 = 10**

1. Choose the correct alternatives for the following:
  - i. Circuit switching takes place at the ..... layer. **[CO1, Understanding]**  
 a) transport      ☒ b) data link      c) physical      d) none of these
  - ii. The hamming distance  $d(000, 011)$  is - **[CO3, Applying]**  
 a) 0      b) 1      ☒ c) 2      d) none of these.
  - iii. The address space of IPv4 is - **[CO4, Understanding]**  
 a) 0      b) infinite      ☒ c)  $2^{32}$       d) none of these.
  - iv. 3-way handshaking of connection establishment is associated with **[CO6, Remembering]**  
 a) HTTP protocol      b) UDP protocol      c) TCP protocol      ☒ d) FTP protocol.
  - v. Identify the class of IP address 191.1.2.3. **[CO4, Applying]**  
☒ a) Class A      b) Class B      c) Class C      d) Class D.
  - vi. In Go\_Back\_N ARQ, if frames 4, 5, 6 are received successfully then the receiver can send an ACK ..... to the sender. **[CO3, Understanding]**  
 a) 5      ☒ b) 6      c) 7      d) any of these.
  - vii. The total number of links required to connect  $n$  devices using Mesh topology is **[CO2, Understanding]**  
 a)  $2n$       b)  $n(n+1)/2$       c)  $n(n-1)/2$       ☒ d)  $n^2$ .
  - viii. Flow control is the responsibilities of the **[CO2, Remembering]**  
 a) Data link layer      b) Transport layer      c) Both of these      d) none of these.
  - ix. Which of the following is a valid host for network 192.168.10.32/28? **[CO4, Understanding]**  
☒ a) 192.168.10.39      b) 192.168.10.47      c) 192.168.10.14      d) 192.168.10.54.
  - x. Pure ALOHA has a maximum efficiency of **[CO4, Understanding]**  
 a) 18%      ☒ b) 37%      c) 10%      d) none of these.

**Group-B**

**(Answer *any five* questions)**

**6 × 5 = 30**

2. Briefly discuss about the different guided media that are used in computer networks and make a comparison among them. **[CO1, Understanding]** (6)
3. What do you mean by “vulnerable time” in medium access protocols? What is the vulnerable time of pure ALOHA? Justify it. **[CO4, Remembering & Evaluating]** (6)
4. Why is acknowledgment numbered in Stop and Wait protocol? Discuss the situation when unnumbered acknowledgments can create confusion in the sender and receiver end. **[CO3, Analyzing]** (6)

5. What is the advantage of two dimensional parity over simple parity? Explain with suitable example. [CO3, Understanding] (6)
6. An IPv4 packet has arrived with offset value 80 (decimal) and value of HLEN field is 10 (decimal). The value of total length field is 200 (decimal). [CO5, Evaluating]
- What are the first and last byte numbers respectively?
  - What is the length of the option field in header? (3 + 3)
7. Differentiate between Link State and Distance Vector routing algorithms. [CO5, Understanding] (6)

**Group-C**  
(Answer any four questions)

4 × 10 = 40

8. i. A network has subnet mask 255.255.255.224. Determine the maximum or number of Host in this network. Also determine the broadcast address of this network. [CO5, Analyzing]
- ii. Applying CRC algorithm, determine the CRC and the transmitted frame for the bit stream 11010111 and for the generator polynomial  $x^3 + x^2 + 1$ . [CO3, Evaluating] (5+5)
9. i. What is the difference between bit stuffing and character stuffing? [CO2, Understanding]
- ii. "In Go Back N ARQ, sender window size  $> 2^m - 1$ ." Is it correct? Justify. [CO3, Analyzing] (5+5)
10. i. What do you mean by an Autonomous System (AS)? What is the difference between Intra-domain and Inter-domain routings? Give an example of each routing protocol. [CO4, Understanding]
- ii. Explain the two node instability problem in RIP. What is the value of infinity to overcome this problem? [CO5, Remembering] (5+5)
11. i. Compare the TCP header and the UDP header. List the fields in the TCP header that are not part of the UDP header. Give the reason for each missing field. [CO6, Remembering & Understanding]
- ii. TCP opens a connection using an initial sequence number (ISN) of 14,534. The other party opens the connection with an ISN of 21,732. Show the three TCP segments during the connection establishment. [CO6, Analyzing] (5+5)
12. Write short notes on any two of the following: (5+5)
- Network Devices [CO1, Remembering]
  - NAT [CO5, Remembering]
  - IPv6 [CO5, Remembering]

**Aliah University**  
**Even (Spring)-Semester Examination - 2025**  
(For 2<sup>nd</sup> Semester MCA (CSE) Programme )

**Paper Name: Operating System**  
**Paper Code: MCAPGPC09**

**Full Marks: 80**  
**Time: 3 Hrs**

**Group – A**

*Answer any five of the following*

1. Answer briefly in your own word

5 X 2 = 10

- (i) How the problem of external fragmentation can be solved? [CO3, BTL1]
- (ii) How does the system discover thrashing? [CO3, BTL4]
- (iii) Write two differences between logical and physical address. [CO3, BTL1]
- (iv) How does DMA increase system concurrency? [CO4, BTL1]
- (v) Discover the ways to overcome the problems of two – level directory. [CO4, BTL4]
- (vi) Distinguish between batch system and time – sharing system. [CO1, BTL4]
- (vii) What is the main advantage of multiprogramming? [CO1, BTL1]

**Group – B**

*Answer any five of the following*

5 X 5 = 25

- 2. List the steps needed to handle page fault.
- 3. What do you mean by real time system? [CO1, BTL1] Distinguish between hard real time system and soft real time system with suitable example. [CO1, BTL6] 1+2+2
- 4. What is the purpose of system calls? [CO1, BTL1] What are five major activities of OS with regard to process management? [CO1, BTL1] 1 +4
- 5. Discuss the essential properties of the multi-processor system and Distributed system. [CO1, BTL6]
- 6. Distinguish between user-level threads and kernel-level threads? Under what circumstances is one type better than the other? [CO2, BTL4]
- 7. Define scheduler. Explain the functionality of different schedulers in brief. [CO1, BTL1]
- 8. What is main memory? Why a portion of main memory must be of ROM type? Explain in detail. [CO1, BTL5]

**Group – C**

*(Answer any three of the following)*

3 X 15 = 45

- 9.
  - a. State critical section problem? Discuss any two solutions to solve the critical section problem. [CO2, BTL1] 9
  - b. Discuss how the following pairs of scheduling criteria conflict in certain settings. [CO2, BTL5] 6
    - i) CPU utilization and response time
    - ii) Average turnaround time and maximum waiting time
    - iii) I/O device utilization and CPU utilization.
- 10.
  - a. Consider the following page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 How many page faults would occur for all page replacement algorithms, assuming three frames are initially empty? [CO2, BTL3] 9
  - b. How can deadlock be detected? Explain. [CO2, BTL1] 6

11.

- a. Suppose that the following processes arrive for execution at the times indicated. Each process will run the listed amount of time. In answering the questions, use non-preemptive scheduling and base all decisions on the information you have at the time the decision must be made. [CO2, BTL6]

| Process | Arrival Time | Burst Time |
|---------|--------------|------------|
| P1      | 0            | 8          |
| P2      | 3            | 4          |
| P3      | 5            | 1          |

- Find the average turnaround time and average waiting time for these processes with the FCFS and SJF scheduling algorithm? 6
- The SJF algorithm is supposed to improve performance, but notice that we choose to run process P1 at time 0 because we did not know that two shorter processes would arrive soon. What will be the average turnaround time if the CPU is left idle for the first 1 unit and then SJF scheduling is used? Remembering that processes P1 and P2 are waiting during this idle time, so their waiting time may increase. This algorithm could be known as future-knowledge scheduling. 3

- b. Discuss situation under which the most frequently used page replacement algorithm generates fewer page faults than the least frequently used page replacement algorithm. Also discuss under which circumstances the opposite holds. [CO3, BTL6] 6

12.

- What are the functions of Virtual File System (VFS) layer in file system implementation [CO2, BTL1] 3
- If the average page faults service time of 25 ms and a memory access time of 100ns. Calculate the effective access time. [CO5, BTL6] 3
- What are the various disk space allocation methods. Explain in detail. [CO5, BTL1 and BTL5] 9

13.

- Explain about RAID in detail. [CO5, BTL5] 6
- What are the types of path names? Explain with example. [CO4, BTL1] 2
- Define resource allocation graph. Using Banker's algorithm answers the following with information given in below table. [CO1, BTL1 & BTL6] 7
  - How many resources of each type is there?
  - What are the contents of need matrix?
  - Find the safe sequence if exist.

| Process | Max Need |   |   | Allocation |   |   | Available |   |   |
|---------|----------|---|---|------------|---|---|-----------|---|---|
|         | A        | B | C | A          | B | C | A         | B | C |
| P0      | 4        | 3 | 1 | 1          | 0 | 1 | 2         | 1 | 1 |
| P1      | 2        | 1 | 4 | 1          | 1 | 2 |           |   |   |
| P2      | 1        | 3 | 3 | 1          | 0 | 3 |           |   |   |
| P3      | 5        | 4 | 2 | 2          | 0 | 0 |           |   |   |

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## Aliah University

End-Semester Examination (Even Semester) - 2025  
(MCA 1<sup>ST</sup> Year 2<sup>ND</sup> Semester)

Subject Name: Image Processing  
Subject Code: MCAPGPE02

Full Marks: 80  
Time: 3hrs

### Group-A (Answer any two questions)

(5 × 2 = 10)

1.

- Illustrate the term 'Image'. [CO1, BL2]
- Mention the applications of image processing. [CO3, BL3]
- How a digital image can be represented? [CO1, BL2]
- What is meant by image filtering? [CO3, BL3]
- Specify the need for image enhancement. [CO2, BL4]

### Group-B

(Answer any five questions)

(5×6= 30)

- Define Brightness, Hue and Saturation. (2+2+2=6) [CO2, BL3]
- Differentiate Image Enhancement in Spatial domain and in frequency domain. (6) [CO3, BL5]
- What do you mean by Color Model. Explain any CMY color model. (4+5=6) [CO5, BL4]
- Find the number of bits required to store a 256 × 256 image with gray level? Write a short note on neighbors of a pixel. (2+4=6) [CO5, BL4]
- Explain fundamental steps in Digital Image Processing with neat block diagram. (6) [CO2, BL2]
- What are the types of images? Explain types of images based on color. (2+4=6) [CO5, BL4]

### Group-C

(Answer any four questions)

(4 × 10 = 40)

- Define Image Sampling and Quantization. Explain linear stretching methods in details. (4+6=10) [CO3, BL4]
- Write detail note about i) Spatial domain enhancement ii) Frequency domain enhancement. (5+5=10) [CO5, BL4]
- What is the need of Image Transformation? Illustrate the 2D Fourier Transform and its pair. Identify the properties of Fourier Transform. (2+4+4=10) [CO5, BL4]
- State basic criteria for Histogram Equalization. Find out the Probability Density Function for the continuous image having value  $P_r(r) = 2r/(L-1)^2$  where  $0 \leq r \leq L-1$  else 0 otherwise. (3+7=10) [CO2, BL5]
- Write short note. i) FFT ii) Image Segmentation iii) Frequency Domain Filters. (3+3+4=10) [CO4, BL4]